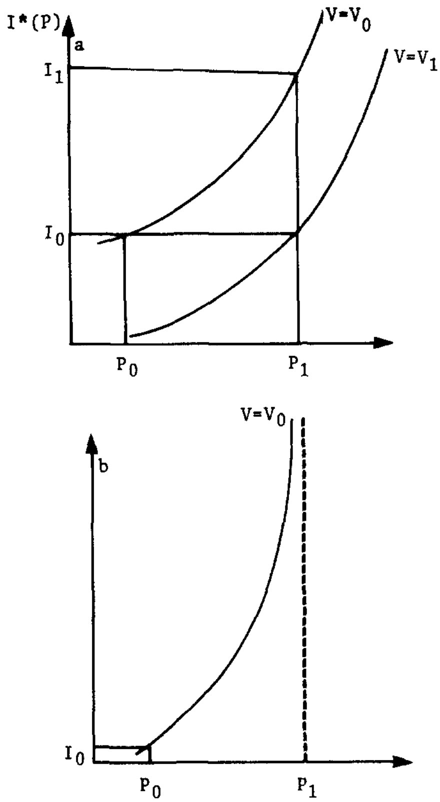

FIG. 1. (a) Indifference curves showing that willingness to pay to avoid increase in pollution from P_0 to P_1 is $I_1 - I_0$. (b) Indifference curve showing that individual cannot be compensated for increase in pollution.

$dI^*(P)/dP$ is the marginal willingness to pay (be compensated) for a marginal decrease (increase) in P. Because V is set equal to V_0, the initial level of utility, in (12), (13) gives the compensating variation. Equivalent variation would be obtained had V been set equal to V_1.

The derivatives of V are easy to compute; because the first-order conditions (5)–(9) hold, many terms drop out. In fact,

$$V_I = \lambda \tag{14}$$

and

$$V_P = (U_S - \lambda w - \lambda m)S_P. \tag{15}$$

But from (7), $U_S - \lambda w - \lambda m = \lambda / S_D$, so that

$$V_P = \lambda \frac{S_P}{S_D} \tag{16}$$

and

$$\frac{dI^*(P)}{dP} = -\frac{V_P}{V_I} = -\frac{S_P}{S_D}. \tag{17}$$

At first glance it would appear that marginal willingness to pay in (17) is independent of personal tastes and income and depends only on the technical characteristics of the sickness (or dose–response) function, S.[7] Note, however, that since D is a choice variable for the individual, the ratio S_P/S_D depends on the characteristics of the utility function and the values of the parameters that the individual faces. S_P/S_D would be independent of D only if $\partial(S_P/S_D)/\partial D = 0$. This condition would be met if S were linear, for example, but is not generally satisfied.

2. TRUE BENEFITS, THE COST-OF-ILLNESS APPROACH, AND DEFENSIVE EXPENDITURES

Let us now examine the relationship between the measure of benefits given in (17) and that provided by the cost-of-illness plus averting or defensive measures. Typically under the COI approach, a dose–response relationship is estimated in a cross-section epidemiological study, like those of Lave and Seskin [9], Crocker [4], or Ostro [11]. This relationship is then used to predict the effect on health status of a change in P. Finally, increases or decreases in S are valued at w to reflect the opportunity cost of illness, and M is linked to sickness through a function like that in (3).

However, virtually no cross-section epidemiological study has included data on defensive expenditures. Since individuals do take such measures to mitigate or prevent the effects of P, what is actually observed in cross-section studies is the total rather than the partial effect of pollution on health. Assuming that the determinants of health other than D and P are invariant with respect to P, the dose–response relationship $S = S(D, P)$ can be totally differentiated to obtain

$$\frac{dS}{dP} = S_D D_P + S_P. \tag{18}$$

Since this total change in health status for a given change in pollution or other insult is used in the COI approach, the added observable cost of

[7]Courant and Porter [3] derive an expression similar to (17) and appear to misinterpret it in a way described in the text.

illness for a change in pollution can be expressed as

$$\frac{dC}{dP} = w\frac{dS}{dP} + m\frac{dS}{dP}, \tag{19}$$

the terms on the right-hand side denoting the lost wages or leisure time due to pollution-related illness, and the resulting out-of-pocket medical costs, respectively.

But from (7),

$$w + m = \frac{U_S}{\lambda} - \frac{1}{S_D}. \tag{20}$$

Combining (18), (19), and (20), we have

$$\frac{dC}{dP} = (w + m)\frac{dS}{dP} = \left[\frac{U_S}{\lambda} - \frac{1}{S_D}\right]\frac{dS}{dP} = \frac{U_S}{\lambda}\frac{dS}{dP} - D_P - \frac{S_P}{S_D}. \tag{21}$$

Hence,

$$\frac{dI^*}{dP} = -\frac{S_P}{S_D} = w\frac{dS}{dP} + m\frac{dS}{dP} - \frac{U_S}{\lambda}\frac{dS}{dP} + D_P. \tag{22}$$

The true willingness to pay to avoid an increase in pollution is, therefore, the amount resulting from the COI approach [the first two terms on the right-hand side of (22)], plus the last two terms. The term $-(U_S/\lambda)(dS/dP)$ is the dollar value of the disutility of pollution-induced illness. D_P is the change in defensive expenditures associated with an increase in pollution.

If $dS/dP > 0$, then marginal willingness to pay always exceeds the sum of changes in defensive expenditures and the cost of illness. Only if this derivative is negative is the sum of the cost of illness and defensive expenditure an overestimate. It is interesting to compare this result with one obtained by Courant and Porter [3], who investigated the suitability of defensive expenditures alone as a measure of the benefits of environmental improvements. They found that defensive expenditures over- or underestimate WTP depending on the shape of the dose–response function, although an underestimate would be the most likely outcome. We agree. According to (22) the only way that defensive expenditures can exceed WTP is for dS/dP to be negative. In other words, individuals would have to respond to an increase in pollution by increasing defensive expenditures so that health improves. This is the only way that the sum of the cost of illness and defensive expenditures can overestimate WTP. Likewise, the only way the cost of illness can overestimate WTP is for D_P to be negative—less than $(U_S/\lambda)(dS/dP)$, in fact. Although nothing in the model prevents either of these outcomes, both would be extremely unlikely in practice.

3. EXTENSIONS OF THE BASIC MODEL

In this section we examine three extensions of the basic model and discuss their effects on benefit estimation. These extensions are as follows:

(i) P is assumed to enter the utility function directly.

(ii) Sickness is assumed to affect the wage rate.

(iii) Individuals are assumed to receive paid sick leave for a fixed work period.

(i) If P has a direct effect on utility, we write the utility function as $U = U(X, L, S, P)$. The first-order conditions (5)–(9) remain unchanged, but the derivative V_P of the indirect utility function becomes

$$V_P = U_S S_P + U_P - \lambda w S_P - \lambda m S_P = \frac{S_P}{S_D} + U_P. \quad (23)$$

If we work through the arithmetic as in the preceding section, we find that

$$\frac{dI^*}{dP} = -\frac{S_P}{S_D} - \frac{U_P}{\lambda} = (w + m)\frac{dS}{dP} + D_P - \frac{U_S}{\lambda}\frac{dS}{dP} - \frac{U_P}{\lambda}. \quad (24)$$

That is, (dI^*/dP) is the same as in our basic case except for the term $-U_P/\lambda$. In this model, the expression for true benefits contains a derivative of U, which is unobservable. Therefore, even if one had data on defensive expenditures, it would not be possible to estimate benefits precisely. Nonetheless, since $U_P < 0$ (no one likes pollution), allowing P to enter the utility function directly makes it even more farfetched for the sum of the cost of illness plus defensive expenditures to exceed willingness to pay. Again if $dS/dP > 0$, then their sum must be an underestimate.

(ii) As Crocker [4] and others have pointed out, it is plausible that the duration of an individual's illness could affect the wage that he or she faces. Chronically ill individuals might be excluded from some physically demanding but high-paying jobs, for example. Or, individuals might be paid a lower wage because of their frequent absences from work.

Suppose this phenomenon is modeled by allowing w to depend on sickness, i.e., $w = w(S)$ where $w'(S) < 0$.[8] This introduces an unexpected twist. The cost of illness can now be written as $C = w(S)S + mS$ which, when differentiated with respect to P, gives

$$dC/dP = [w(S) + Sw'(S) + m]\, dS/dP. \quad (25)$$

Since $w'(S) < 0$, (25) seems to suggest a smaller cost-of-pollution-related

[8]Although it suffices in this simple model to make the wage rate dependent on current illness, a more dynamic formulation in which the wage depends on the worker's cumulative health history would be more realistic.